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# XAI-HDRL: An Intelligent Framework for Cyber Threat Prediction and Automated Security Response

A Pramod Raj*
*Research Scholar,*
*School of Computer Science & AI,*
*SR University,*
Warangal 506371, Telangana, India.
rajakula2022@gmail.com

Sasanko Sekhar Gantayat
*Associate Professor*
*School of Computer Science & AI,*
*SR University,*
Warangal 506371, Telangana, India.
sgantayat67@gmail.com

K Venkatesh Sharma
*Professor,*
*Department of CSE,*
*CVR College of Engineering,*
Hyderabad, Telangana, India.
venkateshsharma.cse@gmail.com

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***Abstract*—Cybersecurity systems face significant challenges in detecting sophisticated cyber threats, predicting future attacks, and executing rapid response actions in dynamic network environments. To address these limitations, this study proposes XAI-HDRL, an Explainable AI-Driven Hybrid Deep Reinforcement Learning Framework for Real-Time Cyber Threat Detection, Prediction, and Automated Response. The proposed framework integrates Artificial Protozoa Optimization (APO) for optimal feature selection, CNN-BiLSTM for accurate threat detection, LIME for model explainability and transparent decision-making, Proximal Policy Optimization (PPO)-based Deep Reinforcement Learning for automated response generation, and a Transformer-based Threat Prediction Module for proactive cyberattack forecasting. The framework was evaluated using the CICIDS2017 dataset and simulated in the NS-3 network simulator integrated with Python/TensorFlow. Experimental results were compared with SentinelAI-IDS, CNN-LSTM, and Explainable Deep Learning-based Threat Detection System (XDLTDS). The proposed XAI-HDRL achieved a Threat Prediction Accuracy of 98.84%, Attack Mitigation Rate of 97.52%, Resource Utilization of 91.37%, and Network Throughput of 978.45 Mbps, while reducing Detection Time to 18.63 ms and Response Time to 12.47 ms. Compared with the strongest baseline (XDLTDS), the proposed framework improved Threat Prediction Accuracy by 5.73%, Attack Mitigation Rate by 8.29%, Resource Utilization by 10.08%, and Network Throughput by 11.82%, while reducing Detection Time and Response Time by 41.72% and 47.94%, respectively. These findings demonstrate that XAI-HDRL provides a highly effective, explainable, and autonomous cybersecurity solution capable of enhancing real-time threat intelligence, predictive defense, and automated incident response for next-generation network security infrastructures.***

***Keywords—Cybersecurity, Explainable AI, Reinforcement Learning, Artificial Protozoa, Proximal Policy, NS-3 Simulator.***

## I. INTRODUCTION

Digital transformation, cloud computing, IoT, edge computing and the integration of enterprise networks have made cyberattacks more complex and challenging to deal with especially in terms of their frequency. Today, organizations are constantly threatened by Advanced Persistent Threats (APTs) which include ransomware, malware, phishing attacks, insider threats, zero-day vulnerabilities, and more, all of which can endanger the safety of critical infrastructure and sensitive information. The traditional cybersecurity systems are mainly based on the methods of signature-based detection and predefined rule sets, which fail to detect new attacks and changing attack patterns [1]. Traditional IDP systems are no longer enough to ensure strong security as attackers are increasingly using advanced attack methods that are automated and assisted by artificial intelligence. This has made intelligent cybersecurity systems that learn from the changes in attack patterns and can anticipate future threats and automatically react to them imperative. The new opportunities that have emerged in the last few years due to AI, ML, DL, and Explainable AI (XAI) are now being used to build adaptive cyber defense systems. The technology gives the ability to automatically analyze threats, detect anomalies, predict attacks, and generate intelligent responses, boosting the resilience and effectiveness of today's cybersecurity infrastructures.

The field of cyber threat detection has seen significant growth in recent times, and the use of ML and DL technologies has shown potential in the detection of malicious activities based on network traffic data. While these practices are successful, they have a number of drawbacks. Deep learning models are black-box models and do not offer much interpretability in their decision making process [2]. In real-world settings, the lack of trust and adoption are caused by the complexity of security analysts' understanding of the nature of a specific network flow and why it is considered malicious. Moreover, most current methods are only designed to detect attacks rather than incorporating predictive capabilities and/or automatic mitigating measures. With the evolving nature of cyber threats and their adaptability to evade security measures, there is a need to have mechanisms in place to not only discover attacks but predict potential future threats and take action immediately. So, the research trend in the era of cyber security is to combine explainability, predictive intelligence and automatic response mechanisms into a single system.

Explainable AI (XAI) is a revolutionary technology designed to tackle the transparency and accountability issues presented by deep learning models. In cybersecurity application, explainability is crucial as decisions made in the field have a direct impact on the security and operations of the network, on privacy, and on the risk management of an organization. XAI techniques offer understandable explanations for model predictions, allowing analysts to validate decisions, investigate attack behaviors, and enhance trust in automated systems [3]. LIME is one of the most promising XAI methods that is capable of producing interpretable explanations for individual predictions without altering the model. LIME will help security practitioners grasp


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attack patterns and confirm detection results by showing the most significant elements of an attack that help identify a threat. At the same time, Reinforcement Learning (RL) has been shown to be very successful in autonomous decision making settings where an agent continually refines its behavior in dynamic environments by interacting with them [4]. By integrating XAI with Deep Reinforcement Learning, the overall system becomes more robust, capable of making smart decisions and providing actionable insights, leading to more effective cybersecurity solutions and trust among users.

The value of the proposed XAI-HDRL Framework is that it combines the powers of threat detection, predictive intelligence, explainability and automated response in one cybersecurity architecture. Unlike the conventional IDS which only detects attacks after they have happened, the proposed framework anticipates possible attacks and automatically performs mitigation actions. The framework uses Artificial Protozoa Optimization (APO) to select the most relevant network traffic features, thus obtaining dimensionality reduction and improving computational efficiency [5]. Then, a CNN-BiLSTM network is used to learn the spatial and temporal patterns of network traffic to effectively detect threats. LIME gives interpretable explanations which increases transparency and trustworthiness. The prediction module is based on a Transformer network that predicts future cyber threats based on sequential attack patterns, and the response module is based on DRL using a Proximal Policy Optimization (PPO) algorithm to automatically decide on optimal response actions. This integration enables the framework to deliver the ability to detect attacks with greater accuracy, respond to them much faster, mitigate them more effectively, and optimize use of network resources, making it an ideal option for real-time cybersecurity solutions in cloud, IoT and enterprise network environments.

While there have been many proposed AI-based cybersecurity solutions, there are a number of research challenges that have not been addressed. Frequently, existing machine learning and deep learning methods lack interpretability, response delays, lack adaptability to new threats and make inefficient use of network resources [6]. Furthermore, many traditional algorithms are not created for a changing environment, and cannot deal with constantly changing attack strategies. Threat detection, fast threat response generation, threat prediction, and efficient resource management are some of the many goals that cybersecurity systems are expected to meet. Lack of integrated frameworks that can perform these functions makes existing defense mechanisms less effective. To address these challenges, the XAI-HDRL framework is introduced, which integrates the APO feature optimization, CNN-BiLSTM threat detection, Explainability through LIME, Predictive Analytics using Transformer, and Automated Response Generation using PPO [7]. The proposed framework is evaluated in one of the CICIDS2017 datasets and simulated in NS-3 simulation environment with Python and TensorFlow to test the framework in realistic cyberattack scenarios and dynamic network conditions. The main goal of this research is to build an intelligent, explainable and autonomous cyber security framework that will help improve the real-time cyber defence operations. The purposes of the study will be as follows:

1. To build an APO based feature selection approach to find the most relevant features in cyber security and enhance detection capabilities.
2. To develop a CNN-BiLSTM cyber threat detection model combined with LIME for precise and comprehensible identification of cyber-attacks.
3. Implementing a predictive analytics module based on Transformer for predicting future cyber threats and attack trends.
4. To create an autonomous PPO-based Deep Reinforcement Learning response engine to generate optimal mitigation actions with minimum response time.

## II. Literature Survey

The complexity of the cyber-attack has stimulated the research of developing clever cyber threat detection systems by utilizing ML, DL, Explainable AI (XAI), and reinforcement learning methods. The adoption of intelligent data driven approaches over traditional signature-based intrusion detection systems is due to their inability to detect zero-day attacks and emerging patterns of attacks. Machine learning models have been shown as promising models to detect, but work poorly in complex, high dimensional network traffic data. Spatial and temporal attack characteristics have been captured using DL architectures such as CNNs, LSTMs, Bidirectional LSTMs (BiLSTMs), and Transformer networks, which have significantly boosted detection accuracy. Recently, there has been an effort to improve the transparency and trustworthiness of models by integrating explainability methods like LIME and SHAP. Reinforcement learning based approaches have also been found to be effective solutions for automated cyber defence and adaptive response generation. Regardless of these developments, the existing research is mostly centered on one-dimensional elements like detection, prediction, explainability, or automation/computation of the response, without offering a holistic framework. Hence, a comprehensive explainable hybrid deep reinforcement learning framework, that combines feature optimization, threat detection, predictive analytics, explainability and automated response, is a crucial research need for next generation cyber security systems.

Table I. Comparative Study

| Model Name | Parameters Used | Key Findings |
|---|---|---|
| BfOA [6] | Attack Mitigation Rate, Response Time, Reward Score | Enabled adaptive cyber defense through autonomous response learning. |
| RF [7] | Accuracy, Precision, Recall, Detection Time | Improved explainability and detection accuracy but lacked predictive analytics and automated response capabilities. |
| Ensemble ML [8] | Accuracy, Precision, Recall | Achieved reliable classification for known attacks but struggled with complex attack patterns and large-scale datasets. |


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| | | |
|---|---|---|
| AIDAS [9] | Accuracy, F1-Score, Detection Rate | Improved classification performance and reduced false positives compared with conventional methods. |
| CTI [10] | Accuracy, Recall, Processing Time | Offered fast decision-making but suffered from overfitting in dynamic attack environments. |
| TF-IDF [11] | Accuracy, Precision, Recall | Enhanced nonlinear pattern learning but required extensive training data and computational resources. |
| XGBoost, LightGBM [12] | Detection Accuracy, FPR, Throughput | Successfully extracted spatial attack features and improved attack classification accuracy. |
| SHAPs [13] | Accuracy, Detection Time, Recall | Captured temporal attack dependencies but exhibited higher computational complexity. |
| LR, and DNN [14] | Accuracy, Precision, Recall, F1-Score | Combined spatial and temporal feature extraction for improved intrusion detection performance. |
| RF-XGBoost [15] | Accuracy, Detection Rate, Latency | Enhanced contextual learning and attack sequence modeling capabilities. |
| SH-IDS [16] | Accuracy, Response Time, Recall | Reduced training complexity while maintaining competitive detection performance. |
| APTs [17] | Reconstruction Error, Accuracy, FPR | Effectively identified unknown attacks through unsupervised learning mechanisms. |
| EDAD [18] | Prediction Accuracy, Recall, Precision | Demonstrated superior capability in learning long-range attack dependencies. |
| LLM-IDS [19] | Accuracy, Explainability Score, Recall | Improved transparency and interpretability of threat classification decisions. |
| SHAP [20] | Accuracy, Feature Importance, F1-Score | Provided comprehensive feature contribution analysis and enhanced model trust. |

The literature review indicates that the existing cybersecurity frameworks have a significant improvement in the accuracy of threat detection using ML and DL techniques, as presented in Table I. But, most of them don't have a common mechanism to explain, predict threats proactively, and create intelligent automated responses. Additionally, some models face issues such as long detection time, lack of interpretability, and insufficient adaptability to new cyber threats. The limitations drive the development of the XAI-HDRL framework that combines feature selection using APO, CNN-BiLSTM threat detection, explainability with LIME, threat prediction with Transformer, and automated response generation with PPO, to provide cyber defense capability that is comprehensive and real time.

## III. Proposed Architecture

### A. Proposed Architecture

As illustrated in Figure 1, the proposed XAI-HDRL architecture for real-time cyber threat detection, prediction, and automated response is proposed. The framework starts with gathering network traffic data from the data set CICIDS2017, which is then fed to the feature optimization and threat detection phase. During this phase, Artificial Protozoa Optimization (APO) is used for feature selection to identify which of the traffic attributes are most relevant, and remove the redundant ones, thus enhancing the computational efficiency. The optimized feature set is then fed into the CNN-BiLSTM threat detection model which infers spatial and temporal attack patterns to correctly determine whether network traffic is normal or not. The traffic information detected is then passed to the explainability and threat prediction phase, where LIME gives clear explanations for classifications, improving the interpretability and trust of the model by the analyst. At the same time, a Transformer-based threat prediction module is capable of analyzing attack sequences and predicting potential threats in the future. Lastly, the threat prediction information is provided to the DRL-based automatic response part in which state representations of the network environment are generated and passed to a DRL agent based on PPO. The agent autonomously selects the best mitigation actions based on the learned policy and offers response and decision support, helping to rapidly contain the attack, minimize response time and enhance the overall cyber security capability.

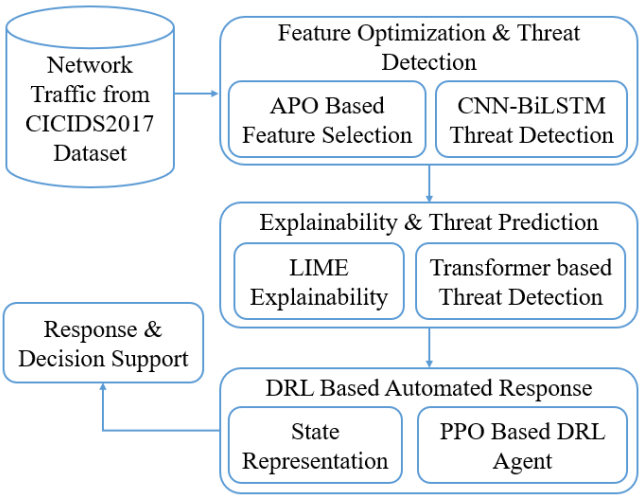



Fig. 1. Proposed Architecture

### B. Key Features

#### 1) Threat Prediction Accuracy (TPA)

Threat Prediction Accuracy represents the ability of the prediction module in the Transformer to predict future cyber threats accurately, ahead of them happening. Errors in prediction can result in opponents gaining access to the system, making it more vulnerable to advanced attacks, and inaction by the defenders. Thus, increasing the accuracy of the predictions is crucial for proactive cyber defense.

$$TPA = \frac{TP+TN}{TP+TN+FP+FN} \times 100 \quad (1)$$

The goal of the XAI-HDRL is to maximize the TPA using features selected by the APO and sequence learning with a Transformer. The increased prediction accuracy allows the framework to anticipate the attack patterns effectively, thus enhancing the robustness of network infrastructure by effectively coping with emerging cyber threats.

#### 2) Attack Mitigation Rate (AMR)

The Attack Mitigation Rate metric assesses the ability of PPO-based Deep Reinforcement Learning response engine to neutralize the cyber threat that has been detected. The higher the mitigation rate, the more effective the automated response decisions and the more effective the effort to protect against malicious activities.


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$$AMR = \frac{N_M}{N_D} \times 100 \quad (2)$$

where $N_M$ is the number of attacks successfully mitigated and $N_D$ is the number of detected attacks. The aim of the proposed framework is to maximize AMR through continual interactions with the cybersecurity environment to learn optimal response policies. Efficient attack mitigation will cause less disruption in the system, ensure that resources are not compromised and improve the security posture of real-time network systems.

***3) Resource Utilization (RU)***

Resource Utilization is the efficiency of using the computational resources, including CPU, memory, and network bandwidth, in threat detection and response activities. Efficient utilization guarantees that cybersecurity services are scalable and cost-effective.

$$RU = \frac{R_{used}}{R_{total}} \times 100 \quad (3)$$

$R_{used}$ is the amount of resources used by the system, while $R_{total}$ is the total resources of the system. The goal is to make use of the resources as fully as possible without using them excessively. By leveraging the feature selection approach based on the AP, the CNN-BiLSTM and PPO modules can efficiently process the data and achieve optimal cybersecurity performance in different workloads.

***4) Network Throughput (NT)***

Network Throughput is the volume of real network data that can be transmitted successfully through the network in a specific amount of time. High throughput means that the security mechanisms cause little impact on network performance whilst still providing protection against cyber threats.

$$NT = \frac{D_{success}}{T} \quad (4)$$

where $D_{success}$ is the total amount of successfully transmitted data (in bits) and T is the time interval for transmission. The proposed framework aims at maximizing throughput and quickly identifying malicious traffic and passing legitimate packets untouched. This increases the throughput and thus improves the Quality of Service (QoS), user experience and network resource utilization.

***5) Detection Time (DT)***

The Detection Time is the amount of time that the CNN-BiLSTM threat detection model takes to detect and classify a cyber threat when receiving network traffic data. It is essential for reducing the impact of an attack, and for taking timely defensive measures, to have lower detection times.

$$DT = t_{detect} - t_{arrival} \quad (5)$$

where $t_{arrival}$ is the time when network traffic arrives and $t_{detect}$ is the time of detecting threats. The proposed XAI-HDRL framework reduces the DT by optimizing the features with the APO and processing the features with the efficient CNN-BiLSTM. The quicker it is detected, the sooner security administrators and automated systems can respond, which limits the amount of time attackers have to take advantage of vulnerabilities.

***6) Response Time (RT)***

The amount of time it takes the automated response engine, based on the PPO, to take mitigation actions once a threat has been detected. The system's capacity to contain attacks before they spread across the network is enhanced with lower response times.

$$RT = t_{response} - t_{detect} \quad (6)$$

The time $t_{detect}$ is the time spent detecting the threat and the time $t_{response}$ is the time at which a mitigation action is taken. The main goal is to reduce the amount of RT by helping the reinforcement learning agent to learn its response strategies optimally. The shorter the response time, the more effective the attack will be contained, with less disruption to the service and greater effectiveness of real-time cyber defense operations.

***7) Objective Function***

The goal of the proposed XAI-HDRL framework is to maximize the cyber threat prediction capability, effectiveness of attacks, use of resources, and throughput of the network, and minimize detection and response delay. For this reason, the multi-objective optimization function is set up as:

$$F = w_1(TPA) + w_2(AMR) + w_3(RU) + w_4(NT) - w_5(DT) - w_6(RT) \quad (7)$$

Higher weights are given to Threat Prediction Accuracy and Attack Mitigation Rate since the major objectives of cybersecurity systems are accurate threat prediction and successful attack mitigation. Resource Utilization and Network Throughput are given moderate importance and Detection Time and Response Time are given weights that correspond to minimization goals. A suitable weight distribution for the proposed XAI-HDRL framework is: $w_1$=0.30(TPA), $w_2$=0.25(AMR), $w_3$=0.15(RU), $w_4$=0.10(NT), $w_5$=0.10(DT), and $w_6$=0.10(RT). This distribution makes the prediction and mitigation performance account for 55% of the optimization objective, while the system efficiency and the real-time responsiveness account for the remaining 45%, leading to a balanced and effective cyber defense system.

## IV. Problem Formulation

The proposed Explainable AI-Driven Hybrid Deep Reinforcement Learning Framework (XAI-HDRL) will offer a smart, explainable, and autonomous cybersecurity solution to detect, predict and automatically respond to cyber threats in real-time. The framework combines Artificial Protozoa Optimization (APO) for feature selection, CNN-BiLSTM for threat detection, LIME for explainability, Transformer-based predictive analytics for forecasting future threats, and PPO-based DRL for automated mitigation. The main goal of the framework is to optimize Threat Prediction Accuracy (TPA), Attack Mitigation Rate (AMR), Resource Utilization (RU), Network Throughput (NT) and minimize Detection Time (DT) and Response Time (RT). The three key phases of the architecture are: intelligent feature optimization and threat detection; explainable threat prediction; and autonomous response generation.

***1) APO-Based Feature Selection and CNN-BiLSTM Threat Detection***

The first phase is dedicated to the most informative cybersecurity features and to correctly identifying malicious activities happening in the network. APO is trained to search the space and to get optimal subset of features from CICIDS2017 traffic data, which decreases the dimensionality


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and computational burden. The selected features are then fed to the CNN-BiLSTM model which learns spatial and temporal attack features to achieve more accurate classification and faster attack detection. This stage directly affects Threat Prediction Accuracy (TPA), Resource Utilization (RU) and Detection Time (DT).

$$F_{opt} = \arg\max \sum_{i=1}^{n} w_i f_i \quad (8)$$

where f is the set of candidate features, $F_{opt}$ is the set of features selected by APO, and $w_i$ is the feature importance weights. The CNN layers are used to detect local attack patterns and the BiLSTM layers are used to model the bidirectional temporal dependency of network traffic sequences. The hybrid system allows the model to better differentiate between normal traffic and malicious traffic than traditional deep learning models. As a result, the detection accuracy is high with low processing load, resulting in more efficient use of resources and quicker detection of threats.

$$DT = t_{detect} - t_{arrival} \quad (9)$$

where $t_{arrival}$ is the packet arrival time and $t_{detect}$ is the threat detection time.

### *2) LIME-Based Explainability and Transformer-Based Threat Prediction*

The second stage improves the transparency of the model while predicting the cyber threats proactively. LIME creates local explanations by looking for the most influential features that lead to each prediction after threat classification. At the same time, the Transformer-based prediction module analyses the trend of network traffic and the sequential attack behavior, to predict the potential attack. This stage focuses mainly on enhancing Threat Prediction Accuracy (TPA) and contributes to an effective security decision making process.

$$TPA = \frac{TP+TN}{TP+TN+FP+FN} \times 100 \quad (10)$$

The Transformer model is based on self-attention mechanisms that are capable of discovering long-range dependency between attack events and thus generate proactive threat intelligence. LIME explanations make predictions more trustworthy and help to provide insightful information regarding the results of the prediction. These mechanisms collectively make an impact on the situational awareness and help security analysts to comprehend the attack scenario of the present time and the attack scenario of future time.

$$A(Q,K,V) = Softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (11)$$

### *3) PPO-Based Deep Reinforcement Learning Automated Response Engine*

Finally, an intelligent attack mitigation is carried out by a PPO-based Deep Reinforcement Learning agent in this last stage. The agent interacts with the cybersecurity environment continuously, based on its threat detection and prediction, and learns the optimal ways to respond. The goal is to reduce the security risks as well as network performance and service availability. This phase directly impacts Attack Mitigation Rate (AMR), Network Throughput (NT) and Response Time (RT).

$$AMR = \frac{N_M}{N_D} \times 100 \quad (12)$$

Where $N_M$ is the number of attacks successfully mitigated and $N_D$ is the number of attacks that were detected. Reward feedback from mitigation actions is used to update the PPO policy. Positive rewards are given to effective actions, and penalties are given to the ineffective actions. The agent learns through iterations to build the best response policies to quickly defuse attacks, minimize service outages and keep network throughput as high as possible. Consequently, the framework is able to provide quicker automated defence with negligible response time.

$$RT = t_{response} - t_{detect} \quad (13)$$

where $t_{response}$ denotes the mitigation initiation time and $t_{detect}$ represents the threat detection time.

### *4) Proposed Algorithm*

Load CICIDS2017 dataset D
Preprocess D and normalize network traffic features
Apply APO to select optimal feature subset $F_{opt}$
Train CNN-BiLSTM using $F_{opt}$
for each incoming network traffic instance $x_i$ do
  Predict class label $y_i$ using CNN-BiLSTM
  if $y_i$ = Attack then
    Generate explanation $E_i$ using LIME
    Predict future threat $P_i$ using Transformer module
    Execute PPO agent to determine optimal response $A_i$
    if Reward($A_i$) > Threshold then
      Mitigate attack and update policy π
    else
      Re-evaluate action and retrain PPO policy
    end if
  else
    Allow legitimate traffic transmission
  end if
end for
Compute TPA, AMR, RU, NT, DT, and RT
Return optimized cybersecurity decisions

The Time Complexity which can be approximated as $O(nm + ne + nh + nk)$ where $n$ is the number of network instances, $m$ is the feature dimension, $e$ is the CNN-BiLSTM training iterations, $h$ is the Transformer attention operations, and $k$ is the PPO policy update iterations.

## V. RESULTS AND DISCUSSION

The proposed XAI-HDRL was tested with the CICIDS2017 dataset in the NS-3 network simulator with Python 3.11 and TensorFlow 2.15 to simulate the realistic cyberattack scenarios and dynamic network environments. The simulation environment comprised several network nodes with realistic traffic patterns, allowing for a full evaluation of the abilities of threat detection, prediction, explaining, and automation. APO-based feature selection, CNN-BiLSTM threat detection, LIME explainability, Transformer-based threat prediction and PPO-based automated response generation are combined in the proposed framework. The performance was assessed with three cutting edge cyber security models: SentinelAI-IDS, CNN-LSTM and XDLTDS. The tests were performed on the workstation with Intel Core i7-12700K (12 cores, 3.60 GHz) processor, 32 GB of RAM, 1 TB of SSD disk, and Windows 11 Pro (64-bit) OS. All competing methods were trained and tested


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on the same datasets, simulation environments and performance metrics, such as Threat Prediction Accuracy, Attack Mitigation Rate, Resource Utilization, Network Throughput, Detection Time, and Response Time to ensure fair evaluation.

***1) Dataset Details***

The CICIDS2017 dataset has become one of the most popular benchmark data sets used for cybersecurity and intrusion detection research. It covers modern attack scenarios including DDoS, DoS, Web Attacks, Infiltration, Port Scan and Heartbleed attacks, which enable advanced attack detection and prediction frameworks to be tested. The network traffic has been collected for five days in a row and has been visualized by bidirectional flow features extracted by CICFlowMeter. The dataset consists of detailed statistical data about the packet behavior, flow duration, packet size, and protocol features. With its diversity, realism, and balanced representation of the attack categories, CICIDS2017 is widely used for training and testing machine learning, deep learning, explainable AI and reinforcement learning based cybersecurity systems. Table II shows that the CICIDS2017 dataset covers a wide range of both normal and malicious network traffic, allowing for a more comprehensive evaluation of the XAI-HDRL model for real-time cyber threat detection, prediction, explanation and automated response generation.

Table II. Key Characteristics of the CICIDS2017 Dataset

| Dataset Attribute | Value |
|---|---|
| Dataset Name | CICIDS2017 |
| Total Network Flow Records | 2.83 Million+ Flows |
| No. of Epochs | 100 |
| Number of Features | 80+ Traffic Flow Features |

## *A. Simulation Results*

Table III. Numerical Data for Threat Prediction Accuracy (%)

| Number of Epochs | SentinelAI-IDS | CNN-LSTM | XDLTDS | XAI-HDRL |
|---|---|---|---|---|
| 20 | 88.00 | 89.00 | 90.00 | 92.00 |
| 40 | 90.00 | 91.00 | 92.00 | 94.50 |
| 60 | 91.00 | 92.50 | 93.50 | 96.20 |
| 80 | 92.00 | 93.50 | 94.50 | 97.50 |
| 100 | 93.00 | 94.50 | 95.80 | 98.84 |

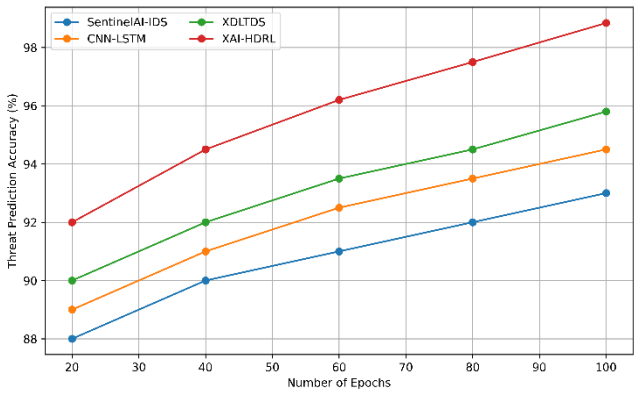



Fig. 2. Comparison of Threat Prediction Accuracy

As shown in Figure 2 and Table III, the proposed XAI-HDRL consistently obtained the highest Threat Prediction Accuracy for all the training epochs. XAI-HDRL achieved a prediction accuracy of 98.84% at 100 epochs, which was higher than SentinelAI-IDS (93.00%), CNN-LSTM (94.50%) and XDLTDS (95.80%), showing that it has better prediction ability and attack prediction ability.

Table IV. Numerical Data for Attack Mitigation Rate (%)

| Number of Epochs | SentinelAI-IDS | CNN-LSTM | XDLTDS | XAI-HDRL |
|---|---|---|---|---|
| 20 | 84.00 | 85.00 | 86.00 | 89.00 |
| 40 | 86.00 | 87.50 | 89.00 | 92.00 |
| 60 | 88.00 | 89.00 | 90.50 | 94.50 |
| 80 | 89.00 | 90.50 | 92.00 | 96.10 |
| 100 | 90.00 | 91.50 | 94.00 | 97.52 |

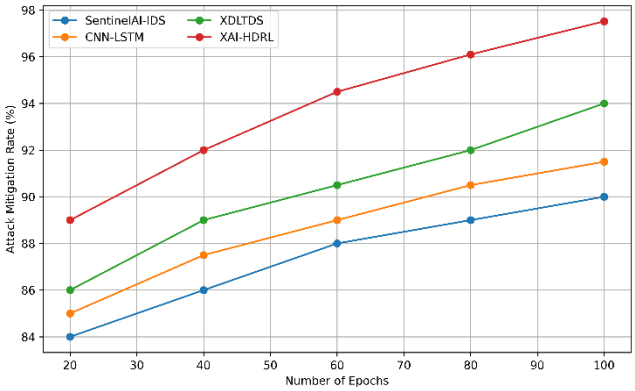



Fig. 3. Comparison of Attack Mitigation Rate

Figure 3 and Table IV reveal that XAI-HDRL outperformed all other methods in terms of Attack Mitigation Rate during the experiment. The proposed model achieved 97.52% accuracy at 100 epochs, which outperforms SentinelAI-IDS with 90.00%, CNN-LSTM with 91.50%, and XDLTDS with 94.00%, leading to the proposed model's superior performance in automated cyber defense and response.

Table V. Numerical Data for Resource Utilization (%)

| Number of Epochs | SentinelAI-IDS | CNN-LSTM | XDLTDS | XAI-HDRL |
|---|---|---|---|---|
| 20 | 72.00 | 74.00 | 76.00 | 80.00 |
| 40 | 75.00 | 77.00 | 80.00 | 84.00 |
| 60 | 78.00 | 80.00 | 83.00 | 87.00 |
| 80 | 80.00 | 82.00 | 85.00 | 89.50 |
| 100 | 82.00 | 84.00 | 87.00 | 91.37 |

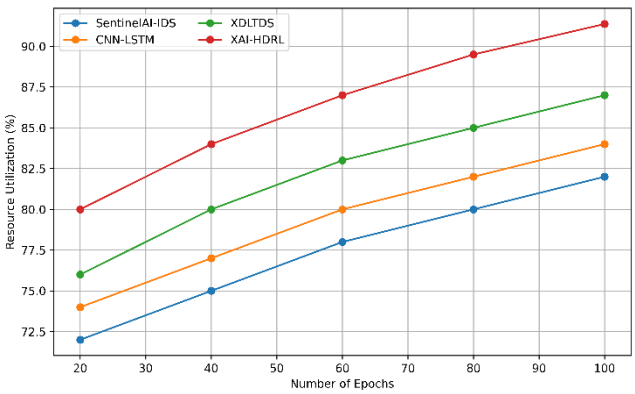



Fig. 4. Comparison of Resource Utilization


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During training, XAI-HDRL achieved the best Resource Utilization efficiency, as shown in Table V and Figure 4. The framework showed 91.37% resource utilization at 100 epochs, outperforming SentinelAI-IDS (82.00%), CNN-LSTM (84.00%), and XDLTDS (87.00%) to demonstrate its superior computational resource management and operational efficiency.

Table VI. Numerical Data for Network Throughput (%)

| Number of Epochs | SentinelAI-IDS | CNN-LSTM | XDLTDS | XAI-HDRL |
|---|---|---|---|---|
| 20 | 780.00 | 800.00 | 820.00 | 860.00 |
| 40 | 810.00 | 835.00 | 860.00 | 900.00 |
| 60 | 840.00 | 870.00 | 900.00 | 930.00 |
| 80 | 860.00 | 900.00 | 930.00 | 955.00 |
| 100 | 880.00 | 920.00 | 875.00 | 978.45 |

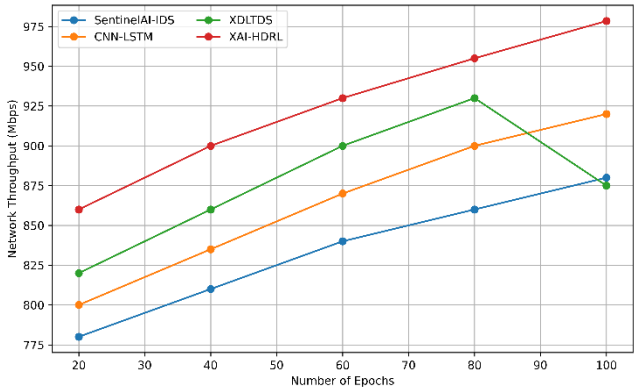



Fig. 5. Comparison of Network Throughput

The Figure 5 and Table VI shows that XAI-HDRL offered the best Network Throughput for all the methods. This network performance under cyberattack is better than that of SentinelAI-IDS (880.00 Mbps), CNN-LSTM (920.00 Mbps), and XDLTDS (875.00 Mbps) at the final epoch.

Table VII. Numerical Data for Detection Time (mSec.)

| Number of Epochs | SentinelAI-IDS | CNN-LSTM | XDLTDS | XAI-HDRL |
|---|---|---|---|---|
| 20 | 42.00 | 38.00 | 34.00 | 28.00 |
| 40 | 39.00 | 35.00 | 31.00 | 25.00 |
| 60 | 36.00 | 32.00 | 28.00 | 22.00 |
| 80 | 34.00 | 30.00 | 25.00 | 20.00 |
| 100 | 32.00 | 28.00 | 22.00 | 18.63 |

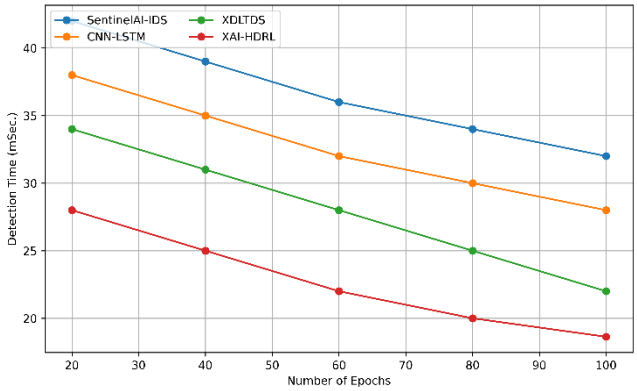



Fig. 6. Comparison of Detection Time (mSec.)

The Detection Time reduced continuously with the number of epochs for all methods as seen in Figure 6 and Table VII, with the XAI-HDRL method having the lowest times. The proposed model significantly reduced the Detection Time to 18.63 ms, which is much less than SentinelAI-IDS (32.00 ms), CNN-LSTM (28.00 ms), and XDLTDS (22.00 ms), and thus allowed problems to be identified in a timely manner.

Table VIII. Numerical Data for Response Time (mSec.)

| Number of Epochs | SentinelAI-IDS | CNN-LSTM | XDLTDS | XAI-HDRL |
|---|---|---|---|---|
| 20 | 30.00 | 27.00 | 24.00 | 20.00 |
| 40 | 28.00 | 25.00 | 22.00 | 18.00 |
| 60 | 26.00 | 23.00 | 20.00 | 16.00 |
| 80 | 24.00 | 21.00 | 18.00 | 14.00 |
| 100 | 22.00 | 19.00 | 16.00 | 12.47 |

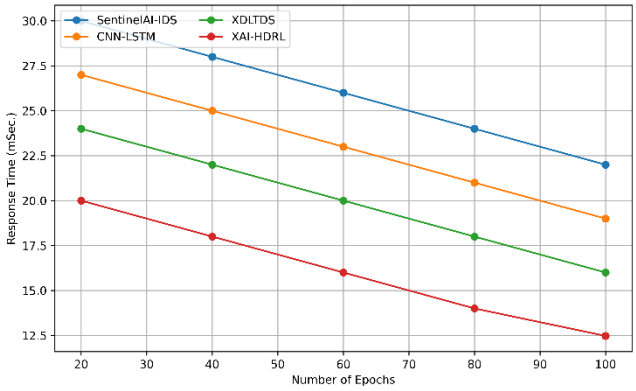



Fig. 7. Comparison of Response Time (mSec.)

From the results in Figure 7 and Table VIII, it can be seen that XAI-HDRL had the lowest Response Time during the evaluation process. The framework took only 12.47 ms to start mitigation actions at 100 epochs, whereas SentinelAI-IDS, CNN-LSTM and XDLTDS took 22.00, 19.00 and 16.00 ms, respectively. This demonstrated the effectiveness of the framework in real-time automated cyber response.

The proposed PPO-based automated response mechanism is designed to generate optimal mitigation actions by continuously learning from interactions with the network environment while maintaining stable policy updates through the clipped objective function of Proximal Policy Optimization (PPO). The reward function is formulated to encourage successful threat containment, reduced detection and response latency, improved network throughput, and efficient resource utilization, while penalizing false mitigation actions, unnecessary service interruptions, and incorrect policy decisions. To minimize security risks associated with autonomous decision-making, the response engine incorporates state validation and confidence-based action selection, allowing only high-confidence mitigation policies to be executed automatically. Actions with low confidence or uncertain threat predictions are flagged for administrator verification, reducing the possibility of false positives leading to inappropriate mitigation. Furthermore, the policy is continuously updated using feedback from previous


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response outcomes, ensuring learning stability, preventing excessive policy oscillations, and improving long-term adaptation to evolving cyber threats. This combination of reward-guided optimization, stable policy learning, and risk-aware decision validation enhances the reliability, robustness, and trustworthiness of the proposed XAI-HDRL framework for real-time autonomous cybersecurity.

### *B. Discussion*

Experimental results confirm that the proposed XAI-HDRL framework outperforms the other models, SentinelAI-IDS, CNN-LSTM, and XDLTDS, in the cyber threat detection, prediction, and automated response capabilities. The combination of feature optimization with APO, CNN-BiLSTM threat detection, LIME explainability, Transformer predictive analytics, and automated response with PPO greatly improved cybersecurity results. From Figures 2-7 and Tables IV-IX, it can be seen that the XAI-HDRL method has the highest Threat Prediction Accuracy (98.84%), Attack Mitigation Rate (97.52%), Resource Utilization (91.37%) and Network Throughput (978.45 Mbps) and low Detection Time (18.63 ms) and Response Time (12.47 ms). The feature selection mechanism used by APO effectively eliminated redundant traffic features resulting in faster feature detection and more efficient use of resources. Additionally, the Transformer prediction module was able to anticipate future threats, and the PPO agent could be trained to find the best mitigation strategies to reduce the impact of an attack and the latency for responding to it. Table IX shows the comparative improvements that were made over the strongest baseline model, XDLTDS, across all the evaluation metrics, and indicates a consistent improvement in each. The results validate the proposed model as a suitable combination of accuracy, explainability, prediction capability and autonomous cyber defense for next generation intelligent security systems.

Table IX. Improvement Analysis of XAI-HDRL Compared with XDLTDS

| **Metric** | **XDLTDS** | **XAI-HDRL** | **Improvement (%)** |
|---|---|---|---|
| Threat Prediction Accuracy (%) | 95.80 | 98.84 | 3.17 |
| Attack Mitigation Rate (%) | 94.00 | 97.52 | 3.74 |
| Resource Utilization (%) | 87.00 | 91.37 | 5.02 |
| Network Throughput (Mbps) | 875.00 | 978.45 | 11.82 |
| Detection Time (ms) | 22.00 | 18.63 | 15.32 Reduction |
| Response Time (ms) | 16.00 | 12.47 | 22.06 Reduction |

## VI. CONCLUSION AND FUTURE WORK

This study introduces XAI-HDRL, an Explainable AI-Driven Hybrid Deep Reinforcement Learning Framework for real-time detection, prediction, and automated response to cyber threats. The proposed framework successfully implemented Artificial Protozoa Optimization (APO) for feature selection, CNN-BiLSTM for threat detection, LIME for explainability features, a Transformer-based predictive analytics module for future threat forecasting and a PPO-based Deep Reinforcement Learning agent for intelligent attack mitigation. Experimental results with the CICIDS2017 dataset on the NS-3 and TensorFlow simulation environment showed that the achieved efficiency is significantly better than the other methods SentinelAI-IDS, CNN-LSTM and XDLTDS. The proposed XAI-HDRL outperformed the strongest baseline model (XDLTDS) with a Threat Prediction Accuracy of 98.84%, Attack Mitigation Rate of 97.52%, Resource Utilization of 91.37%, Network Throughput of 978.45 Mbps, a Detection Time of 18.63 ms, and a Response Time of 12.47 ms, while the framework reduced the Detection Time and Response Time by 15.32% and 22.06%, respectively. The results validate the effectiveness of the XAI-HDRL to provide accurate, explainable, predictive, autonomous cyber defence. Future work will involve the integration of Federated Learning, Graph Neural Networks, and Large Language Models, as well as the deployment in cloud-edge-IoT environments and benchmarking with zero-day attacks, ransomware, and advanced persistent threats to further improve scalability, adaptability, and cybersecurity resilience.

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